

Healthcare Readmission Reduction Analytics Case Study: Identifying Driver Associated with 30-Day Readmissions and Opportunities for Intervention

Business Problem Statement

Hospital leadership is concerned about 30-day hospital readmissions because they may indicate opportunities to improve patient outcomes, care coordination, operational efficiency, and financial performance. The hospital wants to use healthcare data analytics to better understand which patient, clinical, and diagnostic characteristics are associated with readmission and whether these patterns can be used to identify patient groups that may benefit from additional support at or before discharge.

The goal of this analysis is to move from understanding where and among whom readmissions occur to identifying actionable opportunities for targeted intervention.

Because the available data are observational, the analysis will identify factors and patterns associated with readmission rather than establish that a specific factor causes or makes a readmission preventable.

Primary Business Question

How can the hospital use available patient and clinical data to identify patterns associated with 30-day readmissions, stratify patients by readmission risk, and inform targeted interventions at or before discharge?

Business Objectives

Objective 1 — Establish the Readmission Baseline

Measure and describe current 30-day readmission performance.

Establish the overall 30-day readmission rate and examine how readmissions vary across relevant patient, clinical, and diagnostic characteristics.

Objective 2 — Identify Readmission-Associated Factors

Identify patient and clinical characteristics associated with higher readmission rates.

Analyze demographic, clinical, laboratory, and diagnosis-related characteristics to identify patterns associated with 30-day readmission and determine which patient groups carry a greater readmission burden.

Objective 3 — Identify High-Burden Patient and Clinical Groups

Identify populations and clinical characteristics that represent important opportunities for intervention.

Compare readmission patterns across diagnosis categories, clinical characteristics, and other relevant patient segments to identify groups with both elevated readmission burden and meaningful opportunities for improvement.

Objective 4 — Develop an Exploratory Risk Stratification Approach

Evaluate whether predictive analytics can help distinguish patients with different levels of readmission risk.

Use statistical and machine learning approaches to evaluate the ability of available patient characteristics to predict 30-day readmission and explore how patients could be grouped into different levels of risk.

The resulting models are intended as an analytical proof of concept, not as a clinically validated decision-support tool.

Objective 5 — Translate Findings Into Actionable Opportunities

Develop data-informed recommendations for reducing potentially preventable readmissions.

Translate the analytical findings into practical opportunities for targeted intervention, care coordination, discharge planning, and patient follow-up, while recognizing the limitations of the available data.